

• Entropy vs Gini Impurity:

$$H(S) = -P_+ \log P_+ - P_- \log_2 P_-$$

o/p = 3 categories

$$H(S) = -P_{c1} \log_2 P_{c1} - P_{c2} \log_2 P_{c2} - P_{c3} \log_2 P_{c3}$$

$$G \cdot I = 1 - \sum_{i=1}^n (P_i)^2$$

* whenever dataset is small $\rightarrow$ Entropy

large $\rightarrow$ Gini Impurity

Note: In most of the cases or the problem statements we will use Gini Impurity.